

ALNATRON GROUPS

2026 FINAL YEAR PROJECT

TITLES COMPENDIUM

*Electronics · Electrical · Cyber Security · AI / ML / DL / LLM · Biomedical · Automobile & Mechatronics ·
AI & Data Science · CSE / IT (Python-based)*

❖ **200 Real-World, Industry-Relevant Project Titles** ❖

All titles designed to solve current real-world problems

www.alnatron.com

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■ ELECTRONICS

- #01 AI-Powered Smart Street Lighting Using Adaptive Dimming with IoT and Energy Monitoring
- #02 Real-Time Pothole Detection and Road Damage Alert System Using Raspberry Pi and Deep Learning
- #03 Edge AI Garbage Classification System Using MobileNet on Jetson Nano for Smart Waste Management
- #04 IoT-Based Precision Drip Irrigation with Soil Health Monitoring and LLM Crop Advisory via SMS
- #05 Embedded Fire Detection System Using Multi-Sensor Fusion and TinyML for Industrial Safety
- #06 AI-Powered Sign Language to Speech Converter Using CNN and Flex Sensors on Raspberry Pi
- #07 Smart Helmet for Construction Workers with Gas Detection, Fall Alert, and BLE Location Tracking
- #08 Real-Time Vehicle Drowsiness Detection Embedded System Using Eye-Tracking and Alert Mechanism
- #09 IoT-Based Cold Chain Monitoring System for Pharmaceutical Transport with Anomaly Detection
- #10 CNN-Based Fabric Defect Detection System on Embedded FPGA for Textile Industry Automation
- #11 Wearable Air Quality Monitor with Embedded ML Prediction for Urban Pollution Hotspot Mapping
- #12 Smart Water Meter with Leak Detection, Usage Forecasting, and Municipal IoT Dashboard
- #13 Embedded Computer Vision System for Automated PCB Solder Joint Inspection Using YOLO
- #14 LoRaWAN-Based Forest Fire Early Warning Network with Solar-Powered Sensor Nodes
- #15 Smart Attendance and Access Control System Using Face Recognition on Embedded ARM Platform
- #16 IoT-Based Real-Time Bridge Structural Health Monitoring with Vibration and Load Analysis
- #17 Embedded Voice-Controlled Home Automation for Elderly and Differently-Abled Persons with BLE
- #18 AI-Based Livestock Health Monitoring Using Thermal Imaging and Body Temperature Trends on ESP32
- #19 Real-Time Insect Pest Detection in Crops Using Lightweight CNN Deployed on Raspberry Pi Camera
- #20 Smart Parking Slot Management System with Ultrasonic Sensors, IoT Dashboard, and Payment Integration
- #21 Wearable Infant Monitoring Device with Cry Analysis, SpO2, and Parent Alert via Mobile App
- #22 Embedded Machine Learning for Bearing Fault Diagnosis in Industrial Motors via Vibration Signals
- #23 IoT-Based Automated Classroom Energy Management with Occupancy Detection and HVAC Control
- #24 Deep Learning-Based Drowning Detection in Swimming Pools Using Underwater Camera on Edge Device
- #25 AI-Enhanced Smart Mirror for Skin Condition Monitoring Using Image Processing on Raspberry Pi

Total: 25 Project Titles

■ ELECTRICAL

- #01 High-Efficiency Bidirectional DC-DC Converter for Vehicle-to-Grid (V2G) EV Charging Applications
- #02 MPPT-Optimised Single-Phase Grid-Tied Solar Inverter with Anti-Islanding and Harmonic Filtering
- #03 AI-Based Real-Time Fault Detection in Photovoltaic Arrays Using Current-Voltage Curve Analysis
- #04 GaN-Based Interleaved Boost Converter for Ultra-Fast 150 kW EV DC Charging Stations
- #05 Dual Active Bridge DC-DC Converter with Adaptive Phase-Shift Control for Solid-State Transformer
- #06 Wind Energy Harvesting System with MPPT Control Using Permanent Magnet Synchronous Generator
- #07 Smart Battery Management System for LFP EV Packs with SOC/SOH Estimation Using Neural Networks
- #08 Wireless Charging Pad for Electric Two-Wheelers Using LCC-S Resonant Topology and ZVS Control
- #09 Floating Offshore Wind Turbine Grid Integration with Active Power Smoothing and LVRT Capability
- #10 SEPIC-Based Solar-Battery Hybrid Converter for Rural Microgrid with Islanding Protection
- #11 ML-Driven Predictive Energy Management for Grid-Tied Solar-Wind-BESS Hybrid Power Plant
- #12 SiC-MOSFET LLC Resonant Converter for On-Board EV Charger with 98.5% Efficiency Target
- #13 Supercapacitor-Battery Hybrid Energy Storage with Intelligent DC-DC Interface for EV Peak Demand
- #14 Real-Time Power Quality Monitoring and STATCOM-Based Correction for Industrial Loads
- #15 AI-Based Solar Panel Cleaning Robot with Dust Estimation Using Irradiance and Power Deviation
- #16 Dynamic Wireless Power Transfer for E-Rickshaws in Urban Stop-and-Go Traffic Using Road Coils
- #17 Smart Grid-Integrated EV Aggregator with Demand Response and Optimal Scheduling Algorithm
- #18 Partial Shading Resistant PV System with Reconfigurable Array Topology and Global MPPT
- #19 High Gain Quadratic Boost Converter for Low-Voltage PV Panels to 400 V DC Bus Interface
- #20 AI-Enabled Predictive Fault Isolation in Solar Farms Using Drone Thermography and CNN Classification
- #21 Multi-Port DC-DC Converter for Integrating PV, Wind, and Battery in Off-Grid Residential System
- #22 Flyback Converter with Active Clamp for High-Power LED Street Lamp with Autonomous Dimming
- #23 Model Predictive Control for Three-Phase Shunt Active Power Filter in Harmonic-Rich Microgrids
- #24 EV Battery Second-Life Evaluation System with Electrochemical Impedance Spectroscopy and ML
- #25 Tidal Energy Conversion System with Maximum Power Extraction and Grid Synchronisation Control

Total: 25 Project Titles

■ CYBER SECURITY

- #01 Python-Based Phishing URL Detection Engine Using BERT and Real-Time VirusTotal API Integration
- #02 AI-Powered Intrusion Detection System for Home Networks Using XGBoost on Live Packet Captures
- #03 Ransomware Behaviour Analysis Tool with Python Sandbox and YARA Rule Auto-Generation
- #04 Real-Time Credit Card Fraud Detection API Using Isolation Forest and Kafka Stream Processing
- #05 Python Tool for Automated Dark Web Monitoring and Credential Leak Detection for Enterprises
- #06 LLM-Powered Log Analysis Platform for SOC Analysts with Automated MITRE ATT&CK; Mapping
- #07 Zero-Trust Network Policy Enforcer with Python-Based mTLS Certificate Automation for Microservices
- #08 Deepfake Voice Detection System Using Wav2Vec Embeddings and Real-Time Browser Extension
- #09 Python-Based Digital Forensics Toolkit for Android Memory Dump Analysis and Artifact Recovery
- #10 Social Engineering Attack Simulator Using LLM-Generated Spear-Phishing Emails for Security Awareness
- #11 Real-Time API Security Gateway with ML-Based Anomaly Detection for REST/GraphQL Endpoints
- #12 Automated Vulnerability Scanner for Python Web Apps (Flask/Django) with Remediation Suggestions
- #13 DNS Tunnelling Detection Using Entropy Analysis and Bidirectional LSTM on Passive DNS Logs
- #14 Post-Quantum TLS 1.3 Implementation Using Python and Kyber/Dilithium for Secure Communications
- #15 Smart Contract Vulnerability Detector for Ethereum Using Python Static Analysis and GPT-4o
- #16 Python-Based Botnet C2 Traffic Classifier Using Graph Neural Networks on NetFlow Data
- #17 Encrypted Traffic Malware Classification Without Decryption Using TLS Metadata and CNN
- #18 AI-Driven OSINT Aggregator for Threat Intelligence Using Python, Shodan, and Social Media APIs
- #19 Real-Time Insider Threat Detection Using User Behaviour Analytics (UEBA) with Python and ML
- #20 Automated Penetration Testing Report Generator Using Python, Metasploit API, and LLM Summarisation
- #21 IoT Device Fingerprinting and Rogue Device Detection in Enterprise Networks Using Python ML
- #22 Python Tool for Adversarial Prompt Injection Detection in LLM-Based Production Applications
- #23 Supply Chain Attack Detection in Python Package Ecosystem Using Behavioural Diff Analysis
- #24 Real-Time Face Anti-Spoofing System for Biometric Login Using Depth Maps and Liveness Detection
- #25 Cyber Threat Hunting Dashboard with Python, ELK Stack, and Automated Hypothesis Generation via LLM

Total: 25 Project Titles

■ AI / ML / DL / LLM

- #01 Multimodal RAG System for Engineering Document Q&A; Using LLaVA and ChromaDB Vector Store
- #02 LLM-Powered Autonomous Code Review Agent with Security Flaw Detection and Fix Suggestion
- #03 Real-Time Traffic Congestion Prediction for Smart Cities Using Spatial-Temporal Graph Neural Networks
- #04 Personalised Mental Health Chatbot Using Fine-Tuned LLM with CBT Framework and Safety Guardrails
- #05 Federated Learning Framework for Fraud Detection Across Banks Without Sharing Customer Data
- #06 AI-Based Resume Screening and Job Matching Platform Using Sentence-BERT and Bias Mitigation
- #07 Real-Time Misinformation Detection System for Social Media Posts Using LLM and Claim Verification
- #08 Visual Question Answering System for Manufacturing Floor Inspection Using GPT-4o Vision API
- #09 LLM-Based Legal Document Summarisation and Risk Clause Extraction Tool for Indian Courts
- #10 Emotion-Aware AI Tutoring System Using Facial Expression Recognition and Adaptive Content Delivery
- #11 Time Series Anomaly Detection in Industrial IoT Sensor Data Using Transformer Autoencoder
- #12 AI-Powered Agricultural Yield Prediction Using Satellite Imagery, Weather Data, and XGBoost
- #13 End-to-End Tamil/Hindi Speech Recognition System Using Whisper Fine-Tuning for Regional Languages
- #14 Real-Time Object Re-Identification Across Multiple CCTV Cameras Using Siamese Neural Networks
- #15 Graph RAG System for Knowledge Graph-Based Customer Support in Banking Using Neo4j and Llama
- #16 Explainable AI Dashboard for Loan Default Prediction with SHAP Values and Regulatory Compliance
- #17 Diffusion Model-Based Medical Image Synthesis for Rare Disease Dataset Augmentation
- #18 Autonomous AI Agent for E-Commerce Product Research Using LangChain Tools and Web Browsing
- #19 LLM Fine-Tuning Pipeline for Domain-Specific Legal/Medical Q&A; with LoRA and RLHF
- #20 Real-Time Code Vulnerability Detection Using CodeBERT Deployed as VS Code Extension
- #21 Multi-Agent LLM Framework for Automated Scientific Literature Review and Hypothesis Generation
- #22 AI-Based Predictive Maintenance Platform for Manufacturing Using Sensor Fusion and LSTM-CNN
- #23 Deepfake Video Detection Using Spatiotemporal Inconsistency Analysis with CNN-RNN Hybrid
- #24 Personalised News Aggregator Using Reinforcement Learning from Human Feedback (RLHF) Signals
- #25 AI-Powered Supply Chain Risk Prediction Using NLP on News Feeds and Graph-Based Disruption Modelling

Total: 25 Project Titles

■ BIOMEDICAL

- #01 AI-Based Diabetic Retinopathy Grading System Using EfficientNet Deployed on Low-Cost Android Device
- #02 Wearable ECG Patch with On-Device Atrial Fibrillation Detection Using 1D ResNet and BLE Alert
- #03 Non-Invasive Blood Glucose Estimation Using NIR Spectroscopy and LSTM for Continuous Monitoring
- #04 IoT-Enabled Remote Patient Monitoring System with Vital Sign Anomaly Detection for Rural Hospitals
- #05 Deep Learning-Based Lung Cancer Nodule Detection in CT Scans Using 3D U-Net Architecture
- #06 AI-Powered Sepsis Early Warning System Using EHR Data and Gradient Boosting for ICU Patients
- #07 Wearable Seizure Prediction Armband Using Multi-Channel EEG and Spiking Neural Network on MCU
- #08 Federated Learning Model for Breast Cancer Detection Across Hospitals Without Patient Data Sharing
- #09 Smart Pill Dispenser with Medicine Adherence Monitoring, Computer Vision Verification, and IoT Alert
- #10 AI-Based Autism Spectrum Disorder Screening Tool Using Eye-Tracking and Gaze Pattern Analysis
- #11 Real-Time Surgical Instrument Detection and Tracking Using YOLOv9 for Laparoscopic Procedures
- #12 Portable AI-Powered Malaria Detection System from Blood Smear Images for Field Diagnosis
- #13 Mental Health Monitoring Wearable Using Galvanic Skin Response, HRV, and LLM Journaling Companion
- #14 EEG-Based Motor Imagery Brain-Computer Interface for Robotic Arm Control in Paralysed Patients
- #15 AI Dermatology App for Skin Lesion Classification and Melanoma Risk Scoring Using DenseNet
- #16 Continuous Blood Pressure Estimation from Photoplethysmography (PPG) Waveform Using Deep Learning
- #17 Smart Prosthetic Hand with sEMG Intent Detection and Adaptive Grip Force Using Embedded ML
- #18 AI-Based Dental X-Ray Caries Detection and Periodontal Disease Severity Grading System
- #19 Cyber-Secure Medical IoT Gateway for Hospital Networks with ML-Based Anomaly Detection
- #20 LLM-Powered Clinical Decision Support System for Emergency Triage Using Patient History Summarisation
- #21 AI-Based Sleep Apnoea Detection Using Overnight SpO2 and Respiratory Rate from Wrist Wearable
- #22 Deep Learning Model for Predicting ICU Readmission Using Discharge Notes and NLP Feature Extraction
- #23 IoT-Based Post-Operative Recovery Monitoring System with Wound Healing Image Analysis Using CNN
- #24 AI-Powered Drug Interaction Checker Using Knowledge Graph and LLM for Polypharmacy Patients
- #25 Real-Time Fall Detection and Prevention System for Elderly Using Edge AI and Floor Pressure Sensors

Total: 25 Project Titles

■ AUTOMOBILE & MECHATRONICS

- #01 4D Imaging Radar and Camera Fusion for Pedestrian Detection in Fog and Heavy Rain Conditions
- #02 AI-Powered Predictive Maintenance for Commercial Vehicle Fleets Using CAN Bus Data and LSTM
- #03 In-Tyre Sensor-Based Road Surface Classification for Adaptive ABS Using 1D CNN on MCU
- #04 EV Thermal Management System with Reinforcement Learning for Battery-Motor-Cabin Co-Optimisation
- #05 V2X-Based Vulnerable Road User Alert System for Urban Intersections Using C-V2X PC5 Sidelink
- #06 Soft Robotic Gripper with Tactile Sensing Array and Deep RL for Compliant Industrial Pick-and-Place
- #07 AI-Based Autonomous Parking System Using Ultrasonic, Camera Fusion, and MPC Path Planning
- #08 Real-Time Driver Fatigue Monitoring Using Infrared Eye Tracking and Embedded Alert Mechanism
- #09 Intelligent Speed Adaptation System Using GPS Geofencing and Road Sign Recognition for EVs
- #10 AI-Enabled Pothole Avoidance System with Active Suspension Control Using Road Preview Sensors
- #11 Autonomous Mobile Robot for Warehouse Navigation Using SLAM, LiDAR, and Obstacle Avoidance
- #12 Steer-by-Wire System with Force Feedback and Fault-Tolerant Redundancy for Electric Vehicles
- #13 Hybrid Electric Vehicle Energy Management Using Model Predictive Control and Drive Cycle Prediction
- #14 AI-Based Cargo Load Balancing and Overload Detection System for Commercial Trucks Using IoT
- #15 Smart Traffic Signal Controller Using Reinforcement Learning and Real-Time Vehicle Count via CV
- #16 Collaborative Robot (Cobot) Arm with Human Presence Detection and Impedance Safety Control
- #17 Exoskeleton-Assisted Assembly Glove with sEMG Grip Support for Automotive Factory Workers
- #18 Real-Time Engine Knock Detection and Ignition Timing Optimisation Using Vibration Analysis ML
- #19 EV Battery Swap Station Automation with Computer Vision Verification and Robotic Arm Integration
- #20 AI-Based Lane Departure Warning and Steering Correction System Using Monocular Camera on Jetson
- #21 Autonomous Agricultural Robot for Weed Detection and Targeted Herbicide Spraying Using YOLO
- #22 Vehicle-to-Grid (V2G) Bidirectional Charging Scheduler with Grid Demand Response Optimisation
- #23 Smart Tyre Pressure and Temperature Monitoring with Puncture Prediction Using ML on BLE Node
- #24 AI-Powered Emission Monitoring System for Diesel Vehicles Using Exhaust Gas Sensors and ML
- #25 Drone-Based Infrastructure Inspection System with Crack Detection CNN and Automated Reporting

Total: 25 Project Titles

■ AI & DATA SCIENCE

- #01 Real-Time Stock Market Sentiment Analysis Using Twitter/X Streaming API and FinBERT Model
- #02 AI-Based Customer Churn Prediction for Telecom Using XGBoost and Survival Analysis
- #03 Geospatial Crime Hotspot Prediction for Smart Policing Using Historical Data and KDE with DL
- #04 Automated Data Pipeline for E-Commerce Sales Forecasting Using Prophet and LSTM Ensemble
- #05 LLM-Powered Automated EDA (Exploratory Data Analysis) and Insight Generation for CSV Datasets
- #06 Real-Time Air Quality Index Forecasting for Indian Cities Using LSTM and OpenWeatherMap API
- #07 Recommender System for EdTech Platform Using Collaborative Filtering and Knowledge Tracing
- #08 AI-Based Tax Fraud Detection in GST Filings Using Graph Anomaly Detection and Python
- #09 Predictive Analytics Dashboard for Hospital Bed Occupancy Using ML and Power BI Integration
- #10 Social Network Influence Maximisation for Health Campaign Spread Using Graph Neural Networks
- #11 Automated Insurance Claim Damage Assessment from Vehicle Photos Using Vision Transformer
- #12 AI-Powered Hydrological Flood Prediction System Using LSTM and Satellite Rainfall Estimates
- #13 Real-Time Personalised Dynamic Pricing Engine for Ride-Sharing Using Demand Forecasting ML
- #14 NLP-Based Automated Review Summarisation and Fake Review Detection for E-Commerce Platforms
- #15 Data-Driven Energy Consumption Forecasting for Smart Buildings Using Temporal Fusion Transformer
- #16 AI-Based Electoral Sentiment Mapping Using Constituency-Level Social Media Analysis and NLP
- #17 Automated Financial Report Generation from Structured Data Using LLM and Tabular Transformers
- #18 Real-Time Pandemic Outbreak Prediction Using Mobility Data, Genomic Sequences, and SEIR-ML Hybrid
- #19 AI-Powered Skill Gap Analyser for IT Professionals Using Job Posting NLP and Career Path Modelling
- #20 Customer Lifetime Value Prediction for Retail Banks Using Deep Survival Neural Networks
- #21 Real-Time Soil Erosion Risk Mapping Using Remote Sensing Data, DEM, and Random Forest
- #22 AI-Driven Dynamic Curriculum Generator for Competitive Exam Preparation Using Knowledge Graphs
- #23 Automated Pipeline for Detecting Bias in ML Models Across Gender, Age, and Race in HR Systems
- #24 Sports Performance Analytics System for Cricket Using Ball-by-Ball Data and Win Probability Models
- #25 Real-Time Dashboard for Municipal Water Network Leak Detection Using SCADA Data and ML

Total: 25 Project Titles

■ CSE / IT (PYTHON-BASED)

- #01 AI-Powered Automated Code Plagiarism Detection System for Universities Using AST and BERT Embeddings
- #02 Python-Based Real-Time Fake News Detection API with Multilingual Support and Browser Extension
- #03 LLM-Powered Personal Finance Assistant That Analyses Bank Statements and Gives Investment Advice
- #04 Automated Cloud Cost Optimisation System Using Python, AWS Cost Explorer, and ML Forecasting
- #05 AI-Based Student Performance Prediction and Dropout Prevention System for Online Learning Platforms
- #06 Python Tool for Automated Legal Contract Analysis and Obligation Extraction Using LLM
- #07 Real-Time Collaborative Code Editor with AI Pair Programming and Bug Explanation Using GPT API
- #08 Intelligent Chatbot for Government Scheme Discovery and Application Guidance in Regional Languages
- #09 Python-Based Automated Software Testing Framework Using LLM for Test Case Generation from Specs
- #10 AI-Powered E-Learning Content Generator That Creates MCQs, Summaries, and Flashcards from PDFs
- #11 Real-Time Sign Language Interpreter App Using MediaPipe, CNN, and Python for Hearing-Impaired
- #12 Python-Based Microservices Health Monitoring Platform with Predictive Failure Detection Using ML
- #13 Automated Resume Builder and ATS Score Optimiser Using LLM and Real Job Description Analysis
- #14 AI-Powered Meeting Summariser and Action Item Extractor from Audio Recordings Using Whisper + LLM
- #15 Python Backend for Multi-Tenant SaaS Platform with AI-Based Usage Anomaly Detection
- #16 Real-Time Subtitle Generator with Speaker Identification for Video Conferencing Using Whisper
- #17 AI-Powered Smart Search Engine for Internal Enterprise Documents Using Hybrid RAG Architecture
- #18 Automated Social Media Content Scheduler with AI-Generated Captions and Hashtag Optimisation
- #19 Python-Based Incident Management System with LLM-Powered Root Cause Analysis for DevOps Teams
- #20 Real-Time Object Detection App for Visually Impaired Using YOLO, Python, and Audio Description
- #21 AI-Powered Restaurant Menu Recommendation and Diet Planner Using NLP and Nutritional Database
- #22 Automated Python Pipeline for GDPR/DPDP Compliance Auditing of Databases and Data Flows
- #23 Real-Time Emotion Detection in Customer Support Calls Using Wav2Vec and Alert Escalation System
- #24 Python-Based Peer-to-Peer Skill Exchange Platform with AI Matching and Micro-Credentialing
- #25 AI-Powered Code Migration Tool That Converts Legacy COBOL/Java Systems to Python Microservices

Total: 25 Project Titles